After School
MATH LABS

draft concept

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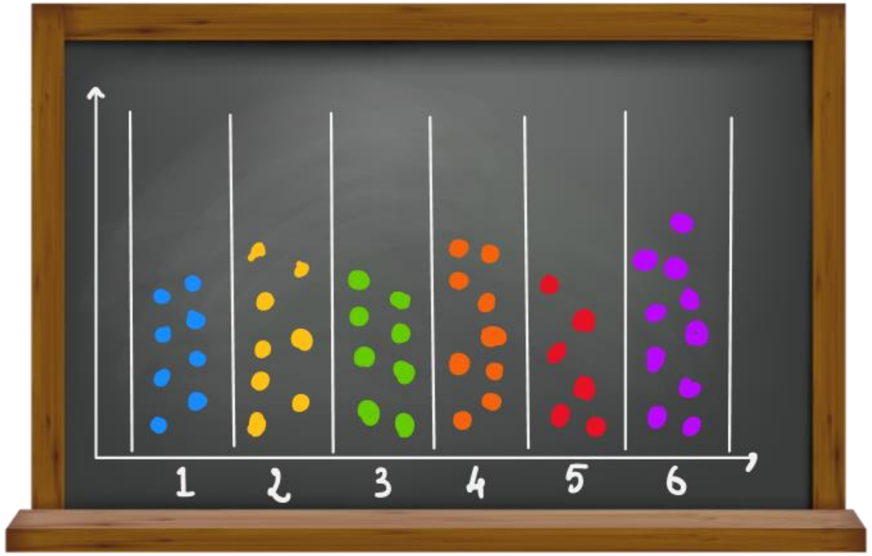
DESIGN PHILOSOPHY

Make it interesting	introduce a stake, a reason for them to care about it
Discuss and predict before experiment	come up with ideas collectively ask: what do we expect? how do you think it will play out
Gamefy	emphasis on collaboration make the task entertaining
Diverse explanation	introduce the concept from multiple angles
Build upon concepts	start with easy, end with difficult

EXAMPLE LAB - PROBABILITY

Combinatorics and Dice Probabilities



Introduce the concept of equally as likely

- start with simple coin toss

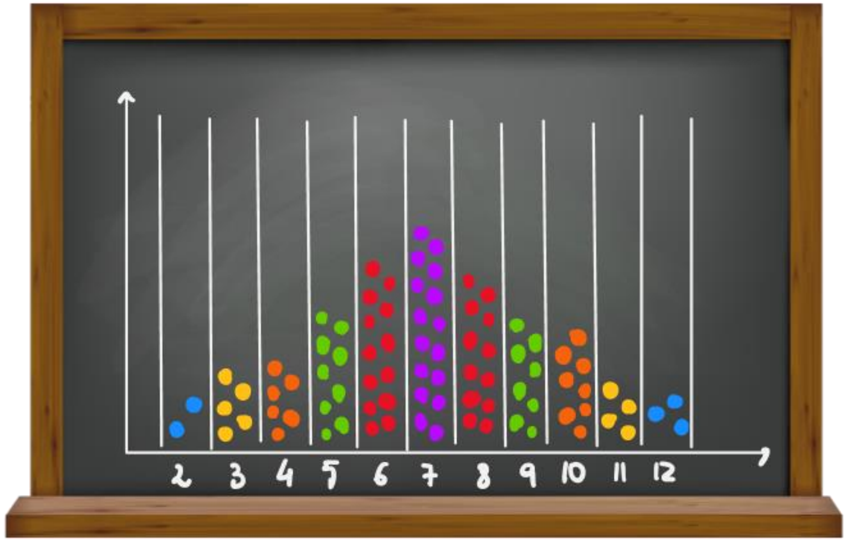
Throw of DICE

- discuss about probability of the numbers

EXPERIMENT

- everyone throws dice and collect the numbers
- create a board with columns: insert a token or draw a point for every time a number comes up
- show that with a lot of repetition all of them have about the same number of points

Combinatorics and Dice Probabilities



Introduce the concept of MORE likely

- bag of marbles with 10 blue and 2 red -> what are we going to draw?

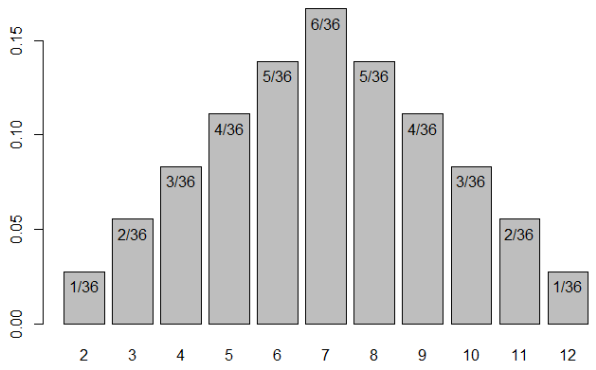
Throw of TWO DICE

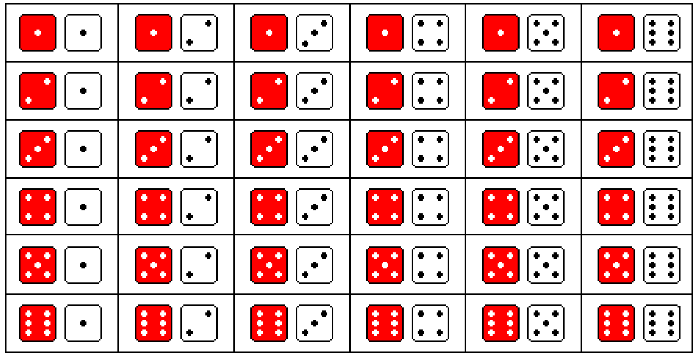
- discuss about the probabilities of each sum combination ->are they the same of is one number more likely?
- draw a board of all 36 combinations

EXPERIMENT

- work in PAIRS: throw one dice each, then sum and collect the numbers
- new board with columns: insert a token or draw a point for every time a summed number comes up
- show that 7 is more likely, 2 and 12 are the least likely

Core Concepts





Probability measure

- Concept of how likely something is

Probability Distribution

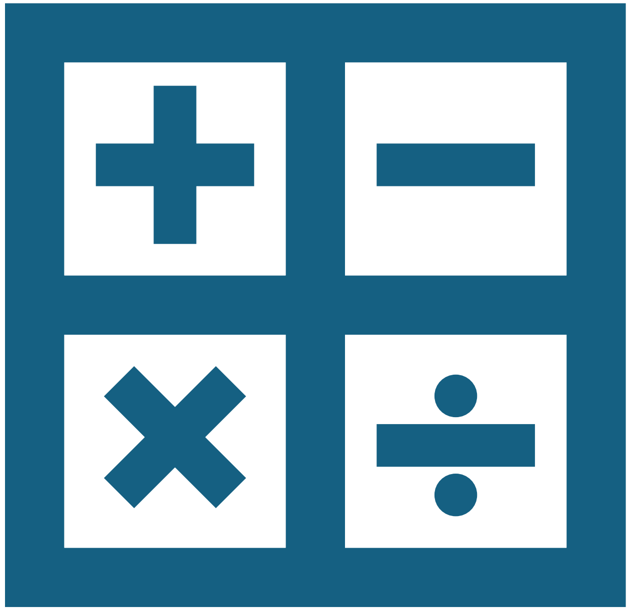
- Columns with points

Random Variable

- Layout of all possible events

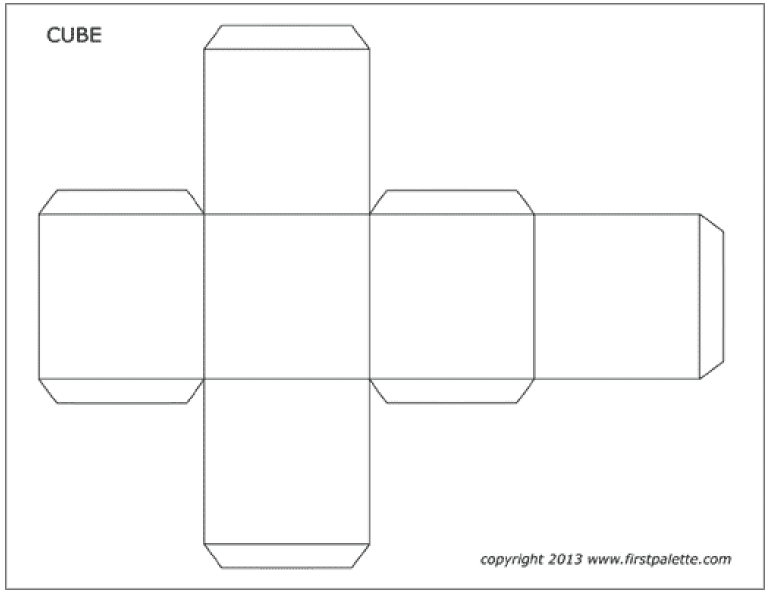
Other Topics

- Geometry
- Concept of a variable
- Possibly Physics:
  - Archimede’s principle
  - Collisions
  - Forces
- Basic Trigonometry
- Cosine and Sine



Possible Modifications

Expand on Concept:	what if we want to see the probability of getting at least 8? Move around the points on the board
Discuss difficulty:	simplified explanation and omission of the all-events chart
More playful/light approach:	if the kids are really young, maybe create the dice together with cut prints of cubes
Different topics	introduce different concepts that teachers may found important/relevant



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